

Name: Team Reef

Date: January 26, 2026

Class: Engineering

MyDesign Portfolio

Element B Initial Template

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Plankton Net>

Source

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some [free online journals and research databases](#)), professional organizations, government agencies, local and national newspapers, patents (here is a [patent search site for the United States Patent and Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image ([Photo 1](#))([Photo 2](#))([Photo 3](#))

Include at least one image of this prior design attempt. Make sure to cite all images.



Details ([Link](#))

Include main details about this prior design attempt

- Cone shaped

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- Fine mesh design
- Made out of nylon or silk with a size of around 80 microns
- Designed to catch microplastics
- Towed by boats

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements. Or describe how you plan to improve on this design.

Unsuccessful:

- Poor filtration efficiency
- Clogging
- Material limitations
- ROV is not being correctly controlled
- Inaccurate volume method

Improvements:

- Optimize mesh size
- Use durable and more useful materials
- Add a flowmeter

Prior Design Attempt #2 - <Sieve>

Source

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Image

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Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

- Mesh screen with very, very small hole sizes to trap small particles but let water and other sediments go through
-

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements. Or describe how you plan to improve on this design.

- “significant reduction in microplastics remaining in the wash water when a dry sieve was implemented immediately after the shredding process.”
- The result was consistent
- It did not get 100% of the microplastics out with only one attempt

Prior Design Attempt #3 - <insert name of prior design attempt #3>

Source

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[Trademark Office](#)), and reputable vendors and manufacturers. Make sure to incorporate [in-text citations](#) in the details section below (i.e. don't just give a list at the end of the element or footnote).



Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

<https://news.cornell.edu/stories/2022/12/students-design-robot-collect-microplastics-beaches>

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements. Or describe how you plan to improve on this design.

- They initially tried to make the robot on a much smaller scale
- Microplastics are pervasive and dangerous
- Existing beach cleanups remove visible trash but often bury microplastics
- Robot features a multi layer filtration system and autonomous docking
- They plan to open source all designs making improvements much easier

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- Their overall mission is to do the greatest good possible and help protect the planet.
- Their testing prototype during 2022 using sand beds, planned beach trials with alumni recommended locations

Prior Design Attempt #4 - <Seabed collection (Grab sampler, box corer, gravity core)>

Source

<https://www.kc-denmark.dk/products/sediment-samplers/box-corer/box-corer-1000-cm%C2%B2.aspx>

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Image



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Is designed for collecting soft sediment samples for lakes or oceans. The design minimises destruction of the sediment.

Include main details about this prior design attempt

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements. Or describe how you plan to improve on this design.

- Issues with the sample disturbance
- Loss of sediment
- Failure to penetrate hard substances
- Disturbed surface structure

Prior Design Attempt #5 - <insert name of prior design attempt #5>

Source

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Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

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Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements. Or describe how you plan to improve on this design.